Chapter 5 — States and Phases of Matter

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Ideal gas equation: (a) PV = nR/T (b) P/V = nRT (c) PT = nRV (d) PV = nRT
2. Gas pressure is due to: (a) Gravity only (b) Bonds (c) Static atoms (d) Wall collisions
3. Hydrogen bonding needs H with: (a) Cl only (b) N, O or F (c) Na or K (d) C or S
4. Ice floats due to: (a) Close packing (b) Ionic lattice (c) Open lattice (d) No H bonds
5. Viscosity decreases with: (a) Cooling (b) More H bonds (c) Temperature rise (d) Larger molecules
6. Evaporation causes: (a) Freezing always (b) Heating (c) Cooling (d) No change
7. Boiling occurs when vapour pressure equals: (a) Zero (b) External pressure (c) Mass (d) Density
8. Liquid crystals are used in: (a) Fuels only (b) Metals only (c) Salts only (d) LCD screens

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why are gases highly compressible?
2. How does kinetic theory explain gas pressure?
3. Why does water have a high boiling point?
4. Why does evaporation produce cooling?
5. Why does boiling point decrease at high altitude?
6. How are liquid crystals useful in displays?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Describe gas properties, the ideal gas equation and kinetic molecular theory. **(4)**
(b) Explain intermolecular forces and hydrogen bonding in water. **(4)**
Q.4 (a) Explain surface tension, viscosity, evaporation, vapour pressure and boiling point. **(4)**
(b) Differentiate crystalline and amorphous solids and describe liquid crystals. **(4)**

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